

11th grade
Term 2 Catch-up Solution: C7 (Confidence Intervals for a Proportion)

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1. Automatic Savings Transfers

Problem description A digital bank wants to estimate the proportion of young adult clients (ages 20–29) who use automatic monthly savings transfers. In a random sample of $n = 120$ clients, $x = 66$ report using automatic transfers. Construct and interpret a 90 % confidence interval for the population proportion.

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We construct a confidence interval for a population proportion:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Step 1: Sample proportion

$$\hat{p} = \frac{x}{n} = \frac{66}{120} = 0,55$$

Check conditions:

$$n\hat{p} = 120(0,55) = 66 \geq 10, \quad n(1 - \hat{p}) = 120(0,45) = 54 \geq 10$$

Step 2: Standard error

$$SE = \sqrt{\frac{0,55(0,45)}{120}} \approx 0,0454$$

Step 3: Margin of error for 90 % confidence ($z^* = 1,645$)

$$E = 1,645(0,0454) \approx 0,0747$$

Step 4: Confidence interval

$$0,55 \pm 0,0747 \Rightarrow [0,475, 0,625]$$

Interpretation: We are 90% confident that between 47.5% and 62.5% of young adult clients use automatic monthly savings transfers.

2. Private Accident Insurance Adoption

Problem description A rideshare platform wants to estimate the proportion of drivers in their 20s who bought private accident insurance this year. In a sample of $n = 200$ drivers, $x = 92$ purchased insurance. Construct 90 %, 95 %, and 99 % confidence intervals for the population proportion, and give a brief interpretation.

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Step 1: Sample proportion

$$\hat{p} = \frac{92}{200} = 0,46$$

Conditions:

$$n\hat{p} = 92 \geq 10, \quad n(1 - \hat{p}) = 108 \geq 10$$

Step 2: Standard error

$$SE = \sqrt{\frac{0,46(0,54)}{200}} \approx 0,0352$$

Step 3: Margins of error

$$E_{90} = 1,645(0,0352) \approx 0,0579$$

$$E_{95} = 1,96(0,0352) \approx 0,0690$$

$$E_{99} = 2,576(0,0352) \approx 0,0907$$

Step 4: Confidence intervals

$$90 \% : [0,46 \pm 0,0579] \Rightarrow [0,402, 0,518]$$

$$95 \% : [0,46 \pm 0,0690] \Rightarrow [0,391, 0,529]$$

$$99 \% : [0,46 \pm 0,0907] \Rightarrow [0,369, 0,551]$$

Interpretation: As confidence increases, the interval becomes wider. All three intervals suggest the insured proportion is around the mid-40 % range.

3. Credit Card Balance Repayment

Problem description A fintech app wants to estimate the proportion of users in their 20s who pay credit card balances in full each month. In a sample of $n = 260$ users, $x = 182$ report paying in full. Construct and interpret a 95 % confidence interval. Then answer: is a benchmark value of $p = 0,75$ plausible based on this interval?

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Step 1: Sample proportion

$$\hat{p} = \frac{182}{260} = 0,70$$

Conditions:

$$n\hat{p} = 182 \geq 10, \quad n(1 - \hat{p}) = 78 \geq 10$$

Step 2: Standard error

$$SE = \sqrt{\frac{0,70(0,30)}{260}} \approx 0,0284$$

Step 3: Margin of error (95 %, $z^* = 1,96$)

$$E = 1,96(0,0284) \approx 0,0557$$

Step 4: Confidence interval

$$0,70 \pm 0,0557 \Rightarrow [0,644, 0,756]$$

Interpretation: We are 95 % confident that between 64.4 % and 75.6 % of similar users pay their balances in full monthly.

Analysis question: Is $p = 0,75$ plausible?

Answer: Yes. The value 0.75 is inside the confidence interval, so it is plausible given this sample at the 95 % confidence level.

4. Monthly Investing Benchmark

Problem description A micro-investing platform is evaluating a support program. It wants the proportion of young clients who invest every month to be at least 60 %. After one cycle, a random sample of $n = 320$ clients finds $x = 173$ monthly investors. Construct a 98 % confidence interval for the population proportion and decide whether the 60 % benchmark is plausible. Justify your decision using the interval bounds.

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Step 1: Sample proportion

$$\hat{p} = \frac{173}{320} = 0,5406$$

Conditions:

$$n\hat{p} = 173 \geq 10, \quad n(1 - \hat{p}) = 147 \geq 10$$

Step 2: Standard error

$$SE = \sqrt{\frac{0,5406(0,4594)}{320}} \approx 0,0279$$

Step 3: Margin of error (98 %, $z^* = 2,326$)

$$E = 2,326(0,0279) \approx 0,0649$$

Step 4: Confidence interval

$$0,5406 \pm 0,0649 \Rightarrow [0,476, 0,606]$$

Interpretation: We are 98 % confident that the true monthly-investor proportion is between 47.6 % and 60.6 %.

Analysis question: Is the benchmark $p \geq 0,60$ plausible, and what decision should the platform make?

Answer: The value 0.60 is inside the interval but very near the upper bound. So 60 % is still plausible, but only marginally. A careful decision is that the program has not clearly established a proportion above 60 %; the platform should continue support and collect more data before claiming the benchmark is comfortably met.

5. Emergency Fund Comparison

Problem description A budgeting startup compares two groups of users in their 20s.

- Group A (users who completed a financial bootcamp): $n = 250$, $x = 170$ keep an emergency fund.
- Group B (users without bootcamp): $n = 250$, $x = 142$ keep an emergency fund.

Construct a 95 % confidence interval for each group proportion. Compare the intervals and discuss what overlap or non-overlap suggests about relative plausibility of higher emergency-fund behavior.

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For each group:

$$\hat{p} \pm 1,96 \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Group A

$$\hat{p}_A = \frac{170}{250} = 0,68$$

$$SE_A = \sqrt{\frac{0,68(0,32)}{250}} \approx 0,0295, \quad E_A = 1,96(0,0295) \approx 0,0577$$

$$CI_{A,95\%} = 0,68 \pm 0,0577 \Rightarrow [0,622, 0,738]$$

Group B

$$\hat{p}_B = \frac{142}{250} = 0,568$$

$$SE_B = \sqrt{\frac{0,568(0,432)}{250}} \approx 0,0313, \quad E_B = 1,96(0,0313) \approx 0,0613$$

$$CI_{B,95\%} = 0,568 \pm 0,0613 \Rightarrow [0,507, 0,629]$$

Interpretation: Group A is estimated around 62.2 % to 73.8 %, while Group B is around 50.7 % to 62.9 %.

Analysis question: What does the overlap pattern suggest?

Answer: The intervals have only a small overlap region (about 62.2 % to 62.9 %). This pattern supports that higher emergency-fund behavior is more plausible in Group A, although there is still some shared plausible range.

6. Retirement Savings by Work Sector

Problem description A national survey studies young workers and compares two sectors.

- Sector 1 (gig logistics): $n = 300$, $x = 126$ contributed to retirement savings last month.
- Sector 2 (junior office jobs): $n = 280$, $x = 151$ contributed last month.

Construct 90 % confidence intervals for both sector proportions. Which sector appears safer in terms of consistently higher retirement-saving participation, and which estimate is more uncertain?

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Sector 1

$$\hat{p}_1 = \frac{126}{300} = 0,42, \quad SE_1 = \sqrt{\frac{0,42(0,58)}{300}} \approx 0,0285$$

$$E_1 = 1,645(0,0285) \approx 0,0469$$

$$CI_{1,90\%} = 0,42 \pm 0,0469 \Rightarrow [0,373, 0,467]$$

Sector 2

$$\hat{p}_2 = \frac{151}{280} = 0,5393, \quad SE_2 = \sqrt{\frac{0,5393(0,4607)}{280}} \approx 0,0298$$

$$E_2 = 1,645(0,0298) \approx 0,0490$$

$$CI_{2,90\%} = 0,5393 \pm 0,0490 \Rightarrow [0,490, 0,588]$$

Interpretation: Sector 2 has a clearly higher interval center and higher bounds.

Analysis questions:

- Which sector appears safer for higher participation?
- Which estimate is more uncertain?

Answers:

- Sector 2 appears safer because its entire interval is shifted upward relative to Sector 1.
- Sector 2 has slightly greater uncertainty here because its interval width (about 0.098) is a bit wider than Sector 1 (about 0.094).

7. Budgeting Rule Use by City

Problem description A payment app tracks budgeting-rule usage (the 50-30-20 rule) among users in their 20s in two cities.

- City A: $n = 360$, $x = 198$ use the rule.
- City B: $n = 340$, $x = 170$ use the rule.

For each city, construct 90 % and 95 % confidence intervals. Then explain how the decision about whether City A is above 55 % changes when the confidence level increases.

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City A

$$\hat{p}_A = \frac{198}{360} = 0,55, \quad SE_A = \sqrt{\frac{0,55(0,45)}{360}} \approx 0,0262$$

$$E_{A,90} = 1,645(0,0262) \approx 0,0431, \quad E_{A,95} = 1,96(0,0262) \approx 0,0514$$

$$CI_{A,90\%} = [0,507, 0,593], \quad CI_{A,95\%} = [0,499, 0,601]$$

City B

$$\hat{p}_B = \frac{170}{340} = 0,50, \quad SE_B = \sqrt{\frac{0,50(0,50)}{340}} \approx 0,0271$$

$$E_{B,90} = 1,645(0,0271) \approx 0,0446, \quad E_{B,95} = 1,96(0,0271) \approx 0,0531$$

$$CI_{B,90\%} = [0,455, 0,545], \quad CI_{B,95\%} = [0,447, 0,553]$$

Interpretation: City A is centered higher than City B at both confidence levels.

Analysis question: How does the decision about $p_A > 0,55$ change?

Answer: At both confidence levels, 0.55 is inside City A's interval. Moving from 90 % to 95 % widens the interval and makes the claim less sharp, so we still cannot say the true proportion is clearly above 55 %; we can only say values around 55 % remain plausible.

8. ETF Adoption Across Two Periods

Problem description An online broker compares two periods for the same age group.

- Period 1: $n = 420$, $x = 231$ users made at least one ETF purchase.
- Period 2: $n = 410$, $x = 254$ users made at least one ETF purchase.

Construct a 95 % confidence interval for each period. Compare uncertainty and discuss whether adoption appears to have increased.

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Period 1

$$\hat{p}_1 = \frac{231}{420} = 0,55, \quad SE_1 = \sqrt{\frac{0,55(0,45)}{420}} \approx 0,0243$$

$$E_1 = 1,96(0,0243) \approx 0,0476$$

$$CI_{1,95\%} = [0,502, 0,598]$$

Period 2

$$\hat{p}_2 = \frac{254}{410} = 0,6195, \quad SE_2 = \sqrt{\frac{0,6195(0,3805)}{410}} \approx 0,0240$$

$$E_2 = 1,96(0,0240) \approx 0,0470$$

$$CI_{2,95\%} = [0,573, 0,666]$$

Interpretation: Period 2 is centered higher, and both intervals have similar width.

Analysis questions:

- Which period has greater uncertainty?
- Does adoption appear to have increased?

Answers:

- Uncertainty is very similar; Period 1 is slightly wider.
- Yes, adoption appears to have increased because Period 2's interval is mostly above Period 1 and the overlap is small.

9. Debt-Management Training Channels

Problem description A policy team evaluates three training channels that teach debt-management skills to young adults.

- Channel A: $n = 280$, $x = 168$ participants reduced debt for three consecutive months.
- Channel B: $n = 300$, $x = 171$ participants reduced debt for three consecutive months.
- Channel C: $n = 260$, $x = 130$ participants reduced debt for three consecutive months.

Construct 90 % and 99 % confidence intervals for all three channels. Then recommend which channel should be prioritized if the target is to keep the true success proportion around or above 60 %, considering uncertainty.

C7

Channel A

$$\hat{p}_A = \frac{168}{280} = 0,60, \quad SE_A = \sqrt{\frac{0,60(0,40)}{280}} \approx 0,0293$$

$$E_{A,90} = 1,645(0,0293) \approx 0,0482, \quad E_{A,99} = 2,576(0,0293) \approx 0,0755$$

$$CI_{A,90\%} = [0,552, 0,648], \quad CI_{A,99\%} = [0,524, 0,676]$$

Channel B

$$\begin{aligned} \hat{p}_B &= \frac{171}{300} = 0,57, \quad SE_B = \sqrt{\frac{0,57(0,43)}{300}} \approx 0,0286 \\ E_{B,90} &= 1,645(0,0286) \approx 0,0470, \quad E_{B,99} = 2,576(0,0286) \approx 0,0736 \\ CI_{B,90\%} &= [0,523, 0,617], \quad CI_{B,99\%} = [0,496, 0,644] \end{aligned}$$

Channel C

$$\begin{aligned} \hat{p}_C &= \frac{130}{260} = 0,50, \quad SE_C = \sqrt{\frac{0,50(0,50)}{260}} \approx 0,0310 \\ E_{C,90} &= 1,645(0,0310) \approx 0,0510, \quad E_{C,99} = 2,576(0,0310) \approx 0,0799 \\ CI_{C,90\%} &= [0,449, 0,551], \quad CI_{C,99\%} = [0,420, 0,580] \end{aligned}$$

Interpretation: Channel A is centered highest and remains closest to the 60% target even at 99% confidence.

Analysis question: Which channel should be prioritized and why?

Answer: Channel A should be prioritized. Its intervals are consistently higher than the others, and its 90% interval is entirely near or above the target threshold. Channel B is moderate, while Channel C is clearly below target in most plausible values.

10. Savings Habit Program Retention

Problem description A youth-finance nonprofit pilots three versions of a savings habit program.

- Program X: $n = 350$, $x = 221$ participants stayed on a weekly savings plan for 12 weeks.
- Program Y: $n = 330$, $x = 191$ participants stayed on plan.
- Program Z: $n = 340$, $x = 180$ participants stayed on plan.

Construct 95% and 99% confidence intervals for each program. Then give a decision recommendation if funding should prioritize programs with a plausibly highest retention proportion and relatively stable uncertainty.

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Program X

$$\begin{aligned} \hat{p}_X &= \frac{221}{350} = 0,6314, \quad SE_X = \sqrt{\frac{0,6314(0,3686)}{350}} \approx 0,0258 \\ E_{X,95} &= 1,96(0,0258) \approx 0,0506, \quad E_{X,99} = 2,576(0,0258) \approx 0,0665 \\ CI_{X,95\%} &= [0,581, 0,682], \quad CI_{X,99\%} = [0,565, 0,698] \end{aligned}$$

Program Y

$$\hat{p}_Y = \frac{191}{330} = 0,5788, \quad SE_Y = \sqrt{\frac{0,5788(0,4212)}{330}} \approx 0,0272$$

$$E_{Y,95} = 1,96(0,0272) \approx 0,0533, \quad E_{Y,99} = 2,576(0,0272) \approx 0,0701$$

$$CI_{Y,95\%} = [0,525, 0,632], \quad CI_{Y,99\%} = [0,509, 0,649]$$

Program Z

$$\hat{p}_Z = \frac{180}{340} = 0,5294, \quad SE_Z = \sqrt{\frac{0,5294(0,4706)}{340}} \approx 0,0271$$

$$E_{Z,95} = 1,96(0,0271) \approx 0,0531, \quad E_{Z,99} = 2,576(0,0271) \approx 0,0698$$

$$CI_{Z,95\%} = [0,476, 0,582], \quad CI_{Z,99\%} = [0,460, 0,599]$$

Interpretation: Program X has the highest center and higher interval bounds at both confidence levels.

Analysis questions:

- Which program is most plausible as the strongest retention option?
- How does changing from 95 % to 99 % affect the confidence in the recommendation?

Answers:

- Program X is the strongest recommendation because both its 95 % and 99 % intervals are shifted above Y and Z.
- Moving to 99 % widens all intervals, increasing caution. Even so, X remains highest, so the recommendation is robust under a stricter confidence level.

11. Minimal Solutions

Problem 1

$$\hat{p} = 0,55$$

$$E = 0,0747$$

$$CI_{90\%} = [0,475, 0,625]$$

Problem 2

$$\hat{p} = 0,46$$

$$E_{90} = 0,0579, \quad E_{95} = 0,0690, \quad E_{99} = 0,0907$$

$$CI_{90\%} = [0,402, 0,518], \quad CI_{95\%} = [0,391, 0,529], \quad CI_{99\%} = [0,369, 0,551]$$

Problem 3

$$\begin{aligned}\hat{p} &= 0,70 \\ E &= 0,0557 \\ \text{CI}_{95\%} &= [0,644, 0,756]\end{aligned}$$

Problem 4

$$\begin{aligned}\hat{p} &= 0,5406 \\ E &= 0,0649 \\ \text{CI}_{98\%} &= [0,476, 0,606]\end{aligned}$$

Problem 5

Group A

$$\begin{aligned}\hat{p}_A &= 0,68, \quad E_A = 0,0577 \\ \text{CI}_{A,95\%} &= [0,622, 0,738]\end{aligned}$$

Group B

$$\begin{aligned}\hat{p}_B &= 0,568, \quad E_B = 0,0613 \\ \text{CI}_{B,95\%} &= [0,507, 0,629]\end{aligned}$$

Problem 6

Sector 1

$$\begin{aligned}\hat{p}_1 &= 0,42, \quad E_1 = 0,0469 \\ \text{CI}_{1,90\%} &= [0,373, 0,467]\end{aligned}$$

Sector 2

$$\begin{aligned}\hat{p}_2 &= 0,5393, \quad E_2 = 0,0490 \\ \text{CI}_{2,90\%} &= [0,490, 0,588]\end{aligned}$$

Problem 7

City A

$$\begin{aligned}\hat{p}_A &= 0,55, \quad E_{A,90} = 0,0431, \quad E_{A,95} = 0,0514 \\ \text{CI}_{A,90\%} &= [0,507, 0,593], \quad \text{CI}_{A,95\%} = [0,499, 0,601]\end{aligned}$$

City B

$$\begin{aligned}\hat{p}_B &= 0,50, \quad E_{B,90} = 0,0446, \quad E_{B,95} = 0,0531 \\ \text{CI}_{B,90\%} &= [0,455, 0,545], \quad \text{CI}_{B,95\%} = [0,447, 0,553]\end{aligned}$$

Problem 8

Period 1

$$\hat{p}_1 = 0,55, \quad E_1 = 0,0476$$

$$CI_{1,95\%} = [0,502, 0,598]$$

Period 2

$$\hat{p}_2 = 0,6195, \quad E_2 = 0,0470$$

$$CI_{2,95\%} = [0,573, 0,666]$$

Problem 9

Channel A

$$\hat{p}_A = 0,60, \quad E_{A,90} = 0,0482, \quad E_{A,99} = 0,0755$$

$$CI_{A,90\%} = [0,552, 0,648], \quad CI_{A,99\%} = [0,524, 0,676]$$

Channel B

$$\hat{p}_B = 0,57, \quad E_{B,90} = 0,0470, \quad E_{B,99} = 0,0736$$

$$CI_{B,90\%} = [0,523, 0,617], \quad CI_{B,99\%} = [0,496, 0,644]$$

Channel C

$$\hat{p}_C = 0,50, \quad E_{C,90} = 0,0510, \quad E_{C,99} = 0,0799$$

$$CI_{C,90\%} = [0,449, 0,551], \quad CI_{C,99\%} = [0,420, 0,580]$$

Problem 10

Program X

$$\hat{p}_X = 0,6314, \quad E_{X,95} = 0,0506, \quad E_{X,99} = 0,0665$$

$$CI_{X,95\%} = [0,581, 0,682], \quad CI_{X,99\%} = [0,565, 0,698]$$

Program Y

$$\hat{p}_Y = 0,5788, \quad E_{Y,95} = 0,0533, \quad E_{Y,99} = 0,0701$$

$$CI_{Y,95\%} = [0,525, 0,632], \quad CI_{Y,99\%} = [0,509, 0,649]$$

Program Z

$$\hat{p}_Z = 0,5294, \quad E_{Z,95} = 0,0531, \quad E_{Z,99} = 0,0698$$

$$CI_{Z,95\%} = [0,476, 0,582], \quad CI_{Z,99\%} = [0,460, 0,599]$$